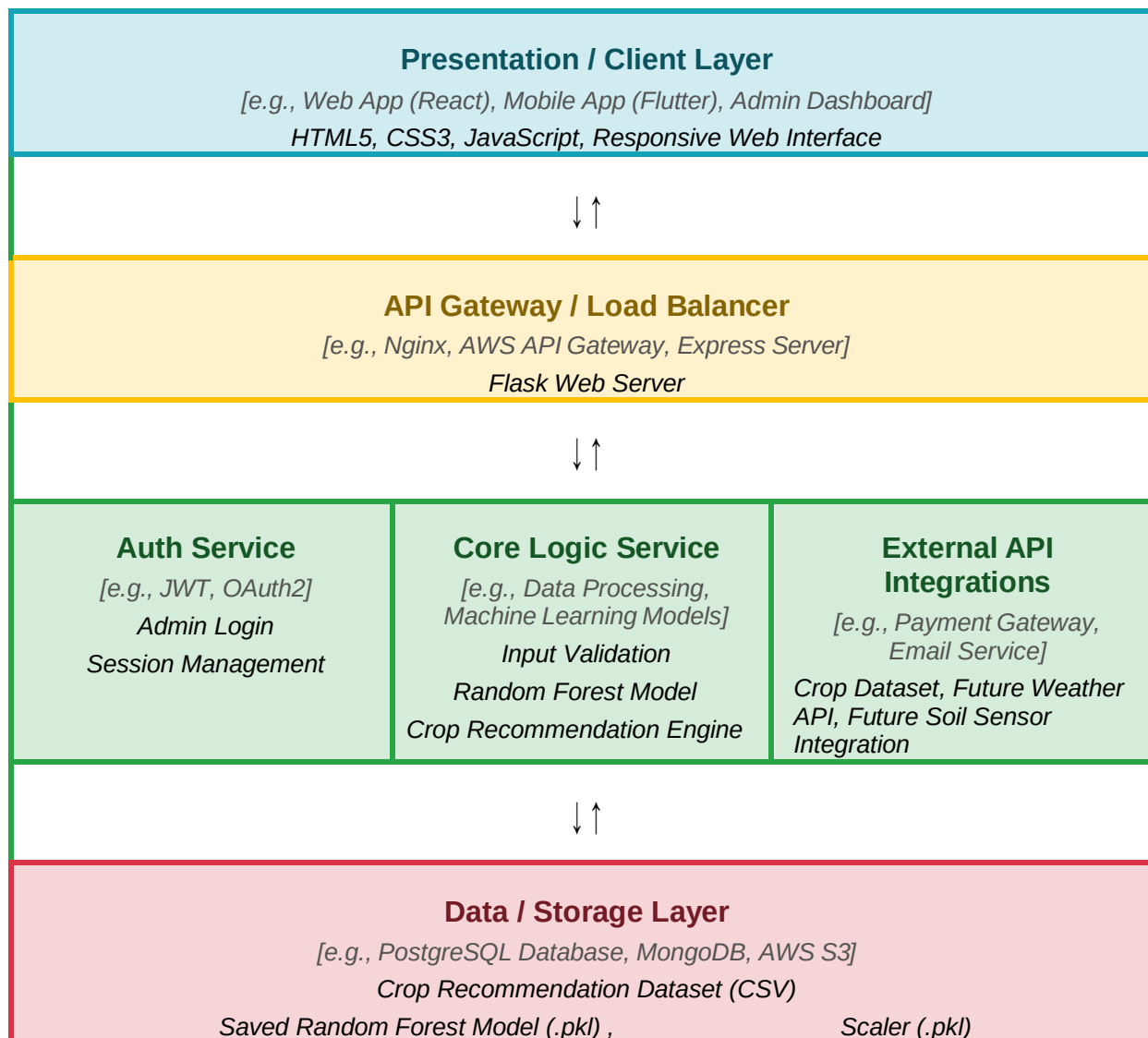


Solution Architecture

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for crop prediction.	HTML, CSS, JavaScript
API Gateway	Handles HTTP requests and connects frontend with backend.	Flask
Core Logic Service	Processes user inputs and predicts the most suitable crop.	Python, Scikit-learn, Random Forest
Database / Storage	Stores dataset and trained models.	CSV Dataset, Pickle Models